ADDIS ABABA UNIVERSITY

SCHOOL OF INFORMATION SYSTEMS

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GROUP MEMBERS

	ID
BLEN TSEGAYE	UGR/6274/17
DAWIT NIGUSSIE	UGR/4605/17
DAGIM YENENEH	UGR/0986/17
ELILTA WORKNEH	UGR/4170/17
MELES DEJENE	UGR/5052/17

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INSTRUCTOR: DEMEKE AYELE

Introduction

The discipline of Information Retrieval (IR) has transitioned from the manual categorization of physical library cards to a sophisticated branch of computational linguistics and data science that underpins the modern digital economy. At its core, Information Retrieval focuses on the identification and extraction of relevant information from vast repositories of unstructured data, a task that has become increasingly complex as the volume of global textual data grows exponentially.

This report presents an exhaustive technical evaluation of a classical Information Retrieval system implemented using the Vector Space Model (VSM) framework, specifically applied to the Cranfield dataset, a foundational benchmark in the field. The system architecture leverages Term Frequency-Inverse Document Frequency (TF-IDF) weighting and Cosine Similarity to facilitate the ranking of document relevance in response to natural language queries.

Dataset Description: The Cranfield Benchmark

The Cranfield dataset is a historic and widely respected benchmark in Information Retrieval, representing the pioneering effort to standardize system evaluation. It was developed specifically to test the efficiency of indexing systems in the aerospace domain.

Structure and Components

The dataset consists of three primary components that together form a complete evaluation framework:

- Documents (cran.all.1400.txt): A collection of 1,400 abstracts from the aerospace engineering literature. The content focuses on high-speed aerodynamics, fluid mechanics, and structural materials.
- Queries (cran.qry.txt): A set of 225 natural language questions reflecting actual information needs formulated by researchers in the 1960s.
- Relevance Judgments (cranqrel.txt): A "gold standard" file mapping each query to the documents deemed relevant by human experts, including a numerical relevance score.

Text processing and Normalization

Preprocessing is the most influential stage in a keyword-based retrieval system, as it directly determines the vocabulary used for indexing. The system applies a rigorous multi-step normalization process

Case Foldng: All text is converted to lowercase, ensuring that "Aerodynamics" and "aerodynamics" are treated as the same token.

Punctuation and Noise Removal: The system uses regular expressions (e.g., re.sub('[^A-Za-z]+', ' ', lines)) to strip away punctuation, numbers, and non-alphabetic characters. This focuses the index on linguistic content rather than experimental data or formatting remnants.

Stopword Removal: Common English words that appear frequently across all documents (e.g., "of," "the," "at") are removed using the NLTK English stopword list. These words have low "resolving power" and their exclusion significantly reduces the dimensionality of the vector space.

Lemmatization: The system applies WordNet-based lemmatization, which uses lexical knowledge and Part-of-Speech (POS) tagging to reduce words to their meaningful dictionary base form (lemma). Unlike rule-based stemming, lemmatization considers the context and grammatical role of a word to ensure accurate normalization. For example, words like *"investigating"* and *"investigation"* are mapped to the base form *"investigate"*, preserving semantic meaning while reducing morphological variation across documents and queries.

Part-of-speech (POS) Tagging: Each token is assigned a grammatical category (noun, verb, adjective, etc.) using POS tagging. This information is used to improve the accuracy of lemmatization by preserving contextual meaning.

Length Filtering: A final regex pass removes very short words (one or two characters), which are typically artifacts of the cleaning process or non-discriminatory abbreviations.[3]

Indexing Approach

The Vector Space Model (VSM) represents each document d_i as a vector in a space of v dimensions, where v is the total number of unique terms in the corpus vocabulary. The value of each dimension corresponds to the weight of that specific term in the document.

The system constructs a Document-Term Matrix D $_{\text{of size}}$ N X V, $_{\text{where}}$ N = 1400. This matrix is initially populated with zeros and subsequently updated with TF-IDF weights. This mathematical transformation allows for the use of linear algebra to calculate similarities between documents and queries, which are also projected into this same V-dimensional space.

Weighting Mechanism

To represent the importance of a term within a document relative to the whole corpus the system employs Term Frequency-Inverse Document Frequency (TF-IDF) weighting. This scheme balances two competing factors.

Term Frequency (TF): measures how frequently the term t occurs in a document d.

$$\text{TF}(t, d) = \frac{\text{Number of times term t appears in document d}}{\text{Total number of terms in document d}}$$

Inverse Document Frequency (IDF): This measures how rare a term is across the entire collection. Terms that appear in nearly every document are less useful for distinguishing relevance than rare technical terms.

$$\text{IDF}(t, D) = \log \frac{\text{Total number of documents in corpus D}}{\text{Number of documents containing term t}}$$

Similarity Measurement

This step involves calculating the similarity between a query vector Q and each document D_i. The system uses Cosine Similarity, which calculates the cosine of the angle between two vectors.

$$\text{cosine_sim}(Q, D_i) = \frac{Q \cdot D_i}{\|Q\| \|D_i\|}$$

Cosine similarity is the preferred metric for the Vector Space Model because it inherently normalizes for document length. A long document and a short document that discuss the same concepts will have vectors pointing in the same direction, resulting in a high similarity score regardless of their absolute term counts. The system ranks documents in descending order of their similarity scores for each query.

Implementation

The system was developed using Python 3, leveraging standard scientific computing and natural language processing libraries to ensure efficiency, modularity, and robustness.

Libraries Used

NLTK: is used for essential natural language processing tasks such as:

- Stopword removal using the English stopword corpus
- Tokenization and linguistic preprocessing
- Part-of-Speech (POS) tagging
- Lemmatization using WordNet-based lemmatizer

Numpy: is used for efficient numerical computation and linear algebra operations. It plays a key role in:

- Initializing the document-term matrix using np.zeros
- Performing vector operations for TF-IDF representation
- Computing cosine similarity between query and document vectors

Pandas: is used for structured data handling, particularly for:

- Reading relevance judgment files (cranqrel.txt)
- Managing evaluation metrics (precision, recall, F1-score)
- Organizing results in tabular format for analysis and reporting

BeautifulSoup: is used as the primary parser for the SGML-formatted Cranfield dataset. It enables structured extraction of document fields such as:

- Title (`.T`)
- Text/Abstract (`.W`)

Work Flow and Code Architecture

Data Loading → Parsing → Preprocessing → Vocabulary Construction → Weighting (TF-IDF representation) → Query processing → Ranking (Similarity computation) → Evaluation

Evaluation Metrics

Precision: The fraction of retrieved documents that are actually relevant

$$Precision = \frac{\text{Relevant Documents Retrieved}}{\text{Total Documents Retrieved}}$$

Recall: The fraction of all relevant documents in the collection that were successfully retrieved.

$$Recall = \frac{\text{Relevant Documents Retrieved}}{\text{Total Relevant Documents in Collection}}$$

F-Score: The harmonic mean of precision and recall

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Effectiveness and Evaluation.

The results demonstrate that the Vector Space Model is highly effective in retrieving a core set of relevant documents for technical aerospace queries. The system achieved a precision of 0.65 at top-ranked results, which is strong for a specialized engineering dataset. This indicates that most of the top retrieved documents are relevant, confirming that TF-IDF weighting successfully identifies highly discriminative technical terms in the Cranfield abstracts.

Implementation Challenges and Limitations

Despite its overall effectiveness, the system exhibits several limitations typical of the bag-of-words Vector Space Model approach.

Preprocessing Sensitivity

System performance is highly dependent on preprocessing decisions. Early experiments using the Porter Stemmer introduced aggressive word reduction, which sometimes distorted technical terms and reduced retrieval precision in domain-specific aerospace vocabulary.

Conclusion

The Vector Space Model implemented in this project provides a strong baseline for information retrieval in technical domains. It achieves good precision and reasonable recall, making it effective for initial document ranking tasks.

However, its main limitation lies in the absence of semantic understanding, which restricts its ability to retrieve conceptually related but lexically different documents. Overall, the system successfully meets its objectives and demonstrates the effectiveness of classical TF-IDF and cosine similarity methods for structured datasets such as the Cranfield collection.

Reference

https://github.com/samujjwaal/Cranfield-Vector-Space-Model.git